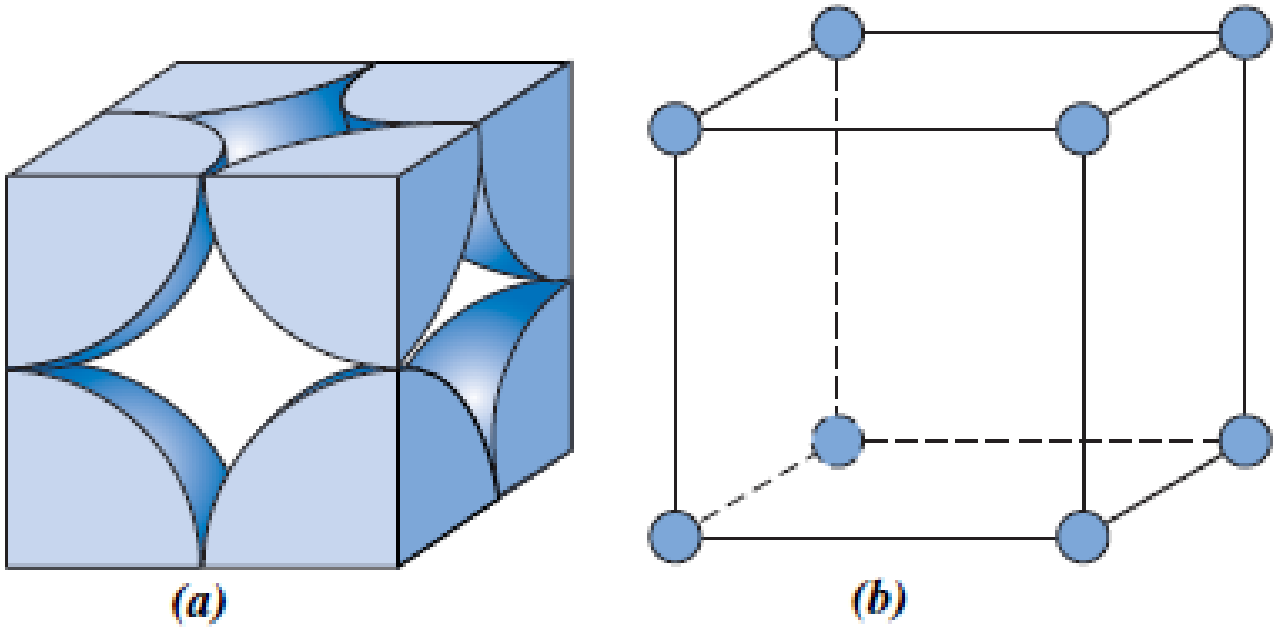
Introduction and Structures of Materials - IIIa



MO-201 Materials Science
Odd Sem 2026-27

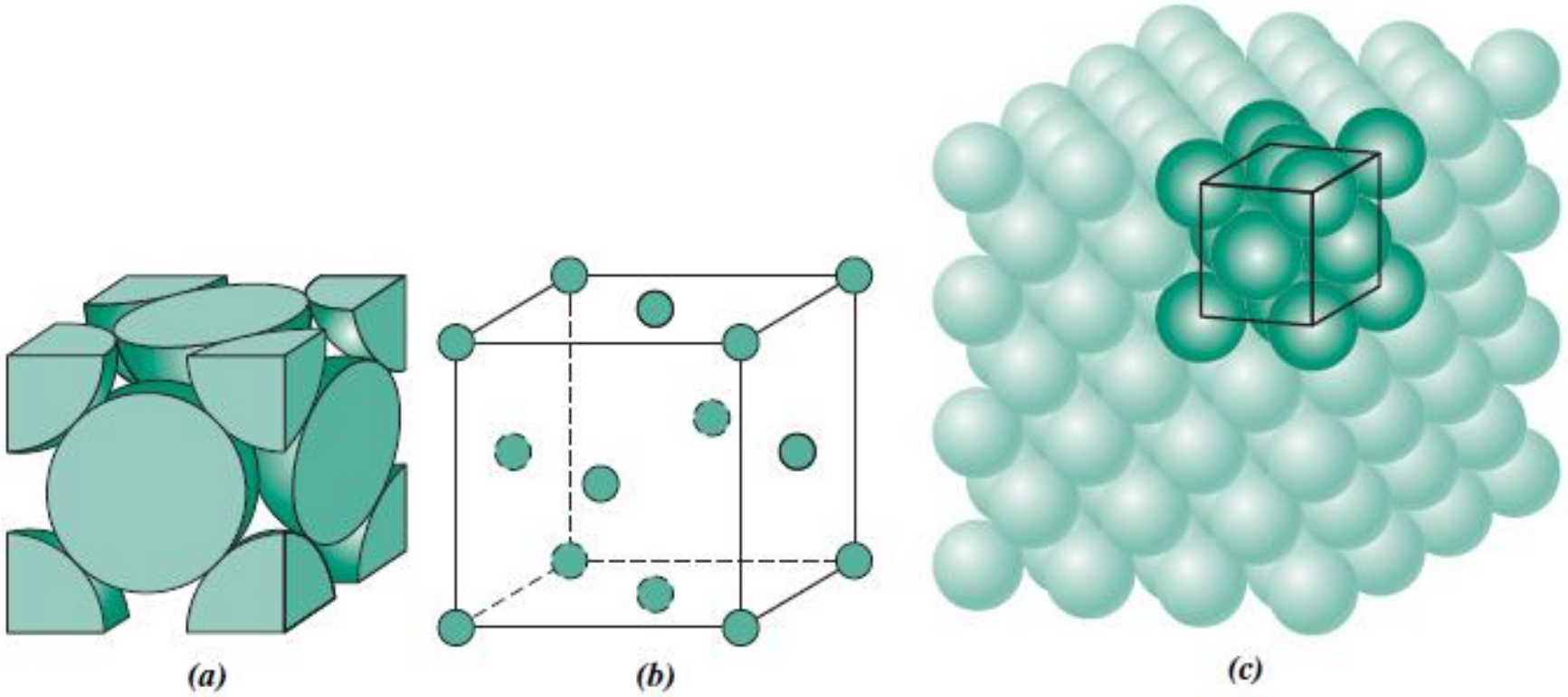
Dr. S. Sinha, Dept. of Metallurgical Engg.
IIT (BHU) Varanasi

Simple Cubic crystal structure



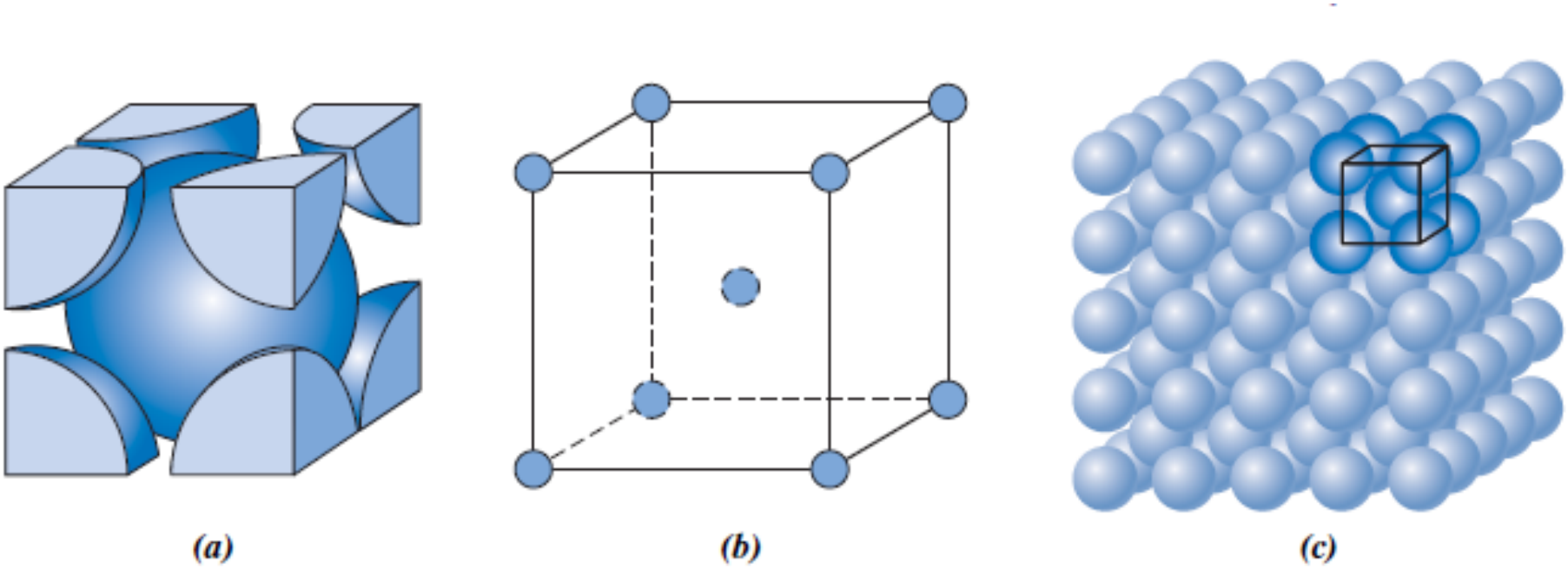
(a) a hard-sphere unit cell, and (b) a reduced-sphere unit cell

Face Centred Cubic crystal structure



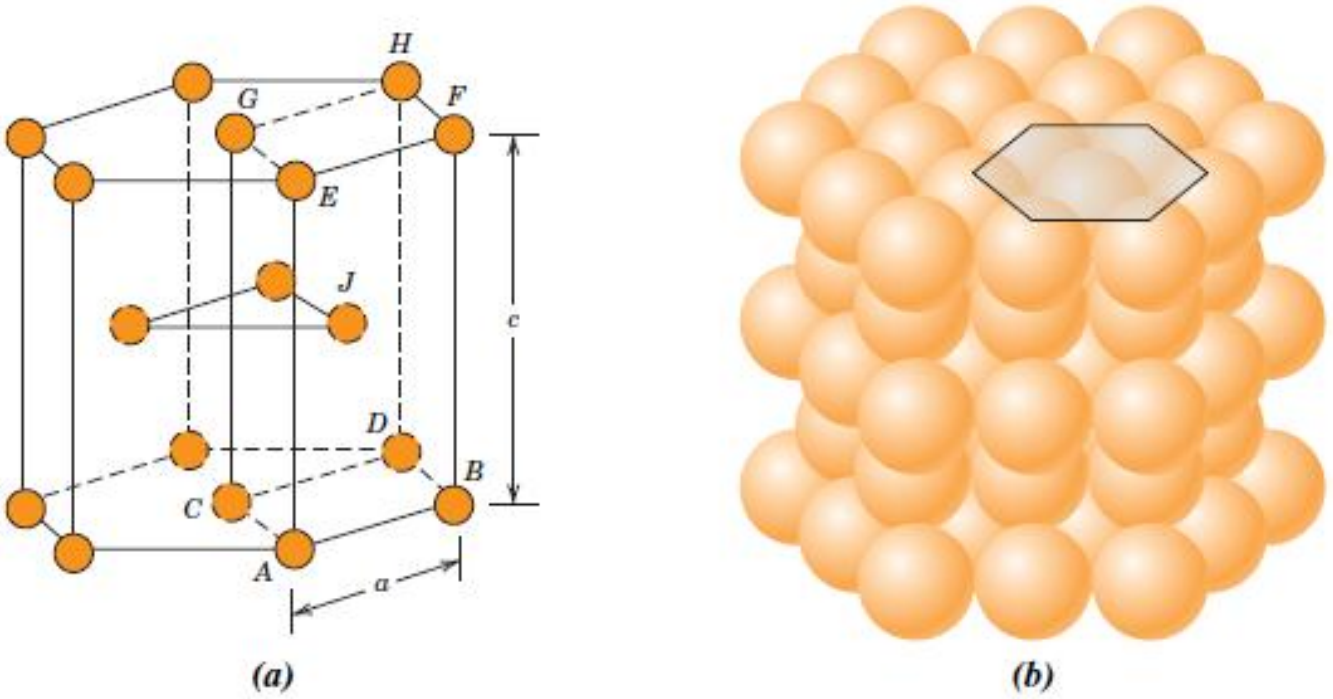
(a) a hard-sphere unit cell representation, (b) a reduced sphere unit cell, and (c) an aggregate of many atoms

Body Centred Cubic crystal structure



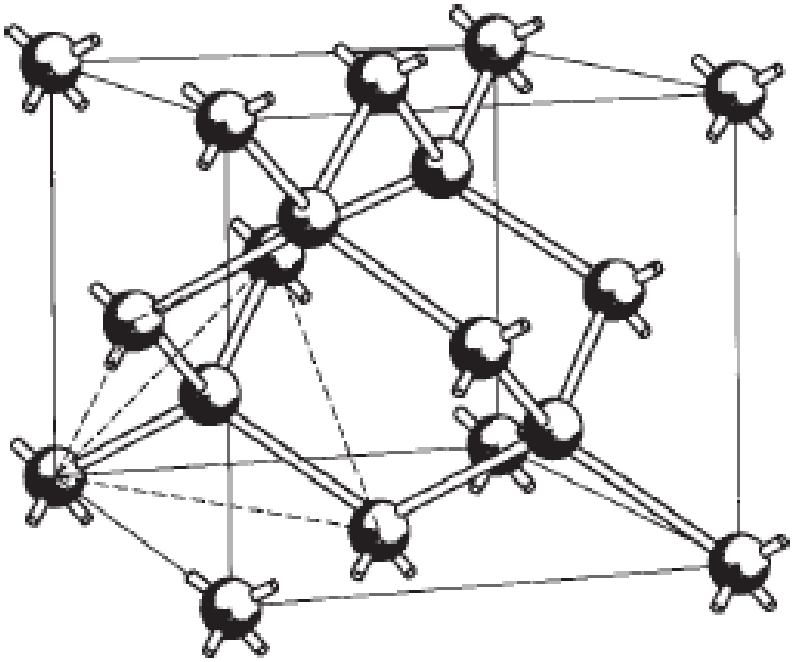
(a) a hard-sphere unit cell representation, (b) a reduced sphere unit cell, and (c) an aggregate of many atoms

HCP crystal structure

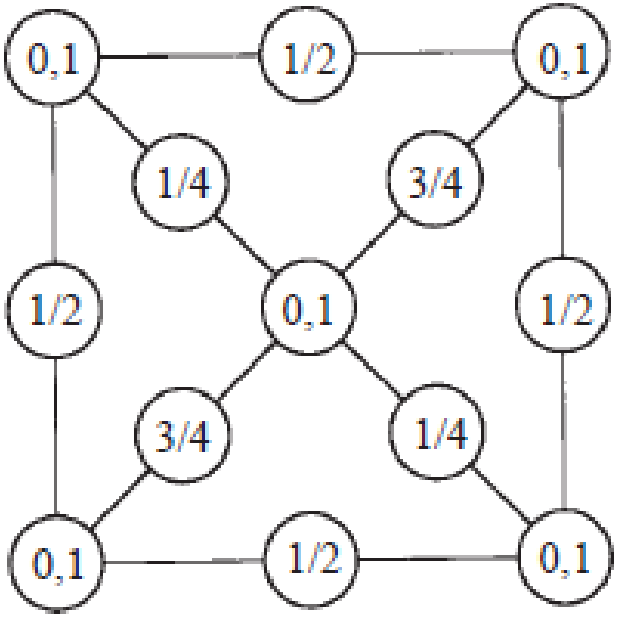


(a) a reduced-sphere unit cell (a and c represent the short and long edge lengths, respectively), and (b) an aggregate of many atoms

Diamond Cubic crystal structure



(a)



(b)

(a) The diamond cubic (DC) unit cell, (b) plan view of atom positions in the unit cell. [The numbers indicate the height from the base of the cell, the total height of the cell being taken as unity.]

Diamond Cubic crystal structure

- The space lattice of the DC crystal is FCC, with two atoms per lattice point
- The basis has one atom at a lattice point, say, at one corner of the unit cell and the other atom at a point quarter way along the body diagonal, which is not a lattice point
- The distance of separation between the two atoms of the basis, which are nearest neighbours, is $a\sqrt{3}/4$, where a is the lattice parameter
- The number of nearest neighbours of a carbon atom, known as the coordination number, is 4
- This low coordination, dictated by the covalent bonding, results in a relatively inefficient packing of the carbon atoms in the crystal.

Bravais Lattices (Space Lattices) contd.

Effective no. of lattice points in unit cell

Space lattice	Abbreviation	Effective number of lattice points in unit cell
Simple cubic	SC	1
Body centred cubic	BCC	2
Face centred cubic	FCC	4

Space lattice	Abbreviation	Effective number of lattice points in unit cell
Simple tetragonal	ST	1
Body centred tetragonal	BCT	2

Space lattice	Abbreviation	Effective number of lattice points in unit cell
Simple orthorhombic	SO	1
End centred orthorhombic	ECO	2
Body centred orthorhombic	BCO	2
Face centred orthorhombic	FCO	4

Bravais Lattices (Space Lattices) contd.

Closest distance of approach

Crystal (monoatomic)	Number of nearest neighbours	Closest distance of approach or atomic diameter
SC	6	a
BCC	8	$a\sqrt{3}/2$
FCC	12	$a/\sqrt{2}$

Coordination number and packing efficiency

Crystal	Coordination number	Packing efficiency
Diamond cubic (DC)	4	0.34
Simple cubic (SC)	6	0.52
Body centred cubic (BCC)	8	0.68
Face centred cubic (FCC)	12	0.74

Example of APF calculation (FCC)

The APF is defined as the fraction of solid sphere volume in a unit cell, or

$$\text{APF} = \frac{\text{total sphere volume}}{\text{total unit cell volume}} = \frac{V_S}{V_C}$$

Both the total atom and unit cell volumes may be calculated in terms of the atomic radius R. The volume for a sphere is $\frac{4}{3}\pi R^3$, and because there are four atoms per FCC unit cell, the total FCC atom (or sphere) volume is

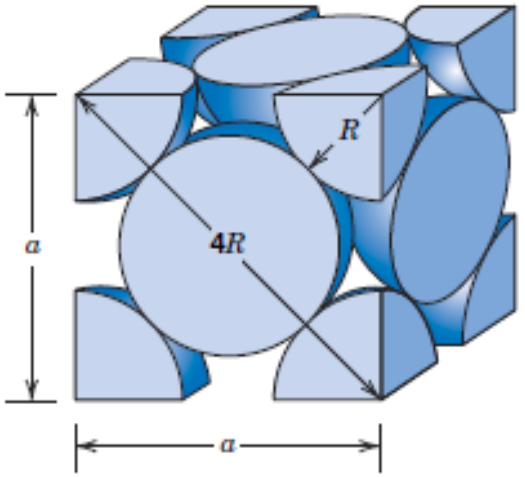
$$V_S = (4)\frac{4}{3}\pi R^3 = \frac{16}{3}\pi R^3$$

In the FCC unit cell illustrated, the atoms touch one another across a face-diagonal, the length of which is $4R$. Because the unit cell is a cube, its volume is a^3, where a is the cell edge length. From the right triangle on the face,

$$a^2 + a^2 = (4R)^2$$

or, solving for a,

$$a = 2R\sqrt{2}$$



The FCC unit cell volume V_C may be computed from

$$V_C = a^3 = (2R\sqrt{2})^3 = 16R^3\sqrt{2}$$

Therefore, the atomic packing factor is

$$\text{APF} = \frac{V_S}{V_C} = \frac{(\frac{16}{3})\pi R^3}{16R^3\sqrt{2}} = 0.74$$

Example of Packing efficiency calculation (Diamond)

Effective number of atoms in the DC unit cell

$$= \frac{1}{8} \times 8 \text{ (corner atoms)}$$

$$+ \frac{1}{2} \times 6 \text{ (face centred atoms)}$$

$$+ 1 \times 4 \text{ (atoms fully within the unit cell)}$$

$$= 8.$$

Volume of each atom $= (4/3)\pi r^3 = (4/3)\pi\ (a\sqrt{3}/8)^3$,

where r is the radius of the atom and a is the lattice parameter.

$$\text{Packing efficiency} = \frac{\text{Volume of atoms in the unit cell}}{\text{Volume of unit cell}}$$

$$= \frac{8 \times (4/3)\pi\ (a\sqrt{3}/8)^3}{a^3}$$

$$= 0.34$$

Example of Packing efficiency calculation (Diamond)

In the diamond cubic unit cell, there are eight corner atoms, six face centred atoms and four more atoms.

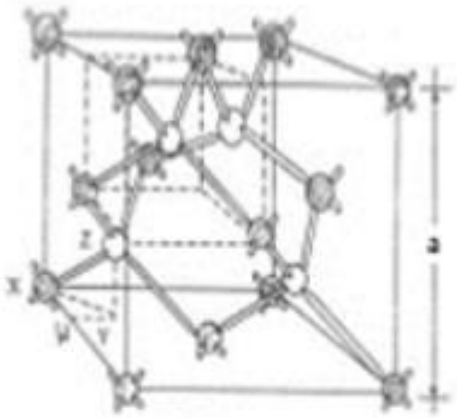
$$xz = 2r = \sqrt{(\frac{a\sqrt{2}}{4})^2 + (\frac{a}{4})^2} = \frac{a\sqrt{3}}{4}$$

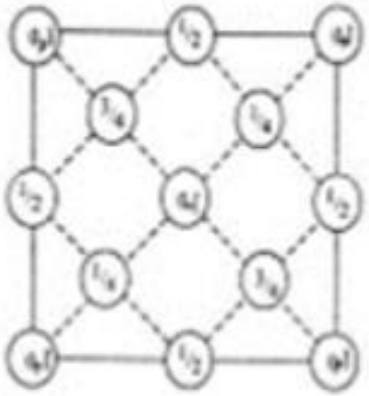
Atomic radius (r) is :

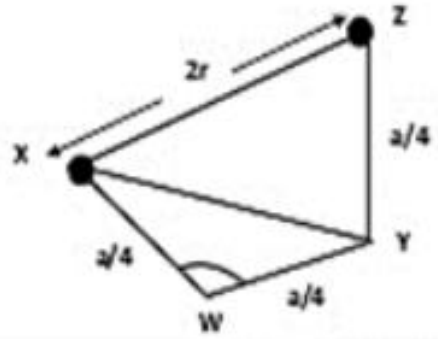
$$r = \frac{a\sqrt{3}}{8}$$

It becomes possible to calculate the APF as follows:

$$APF = \frac{8 \times [\frac{4}{3}\pi(\frac{\sqrt{3}a}{8})^3]}{a^3} = \frac{\pi\sqrt{3}}{16} = 0.34$$







Atomic factor for diamond

Example of Density calculation (Diamond and Cu)

Diamond

$$\text{Density} = \frac{\text{Mass of atoms in unit cell}}{\text{Volume of unit cell}}$$

$$= \frac{12 \times 1.660 \times 10^{-27} \times 8}{(3.57 \times 10^{-10})^3}$$

$$= 3500 \text{ kg m}^{-3}$$

$$= 3.5 \text{ g cm}^{-3}$$

$$\rho = \frac{nA}{V_C N_A}$$

n = number of atoms associated with each unit cell

A = atomic weight

V_C = volume of the unit cell

N_A = Avogadro's number (6.022×10^{23} atoms/mol)

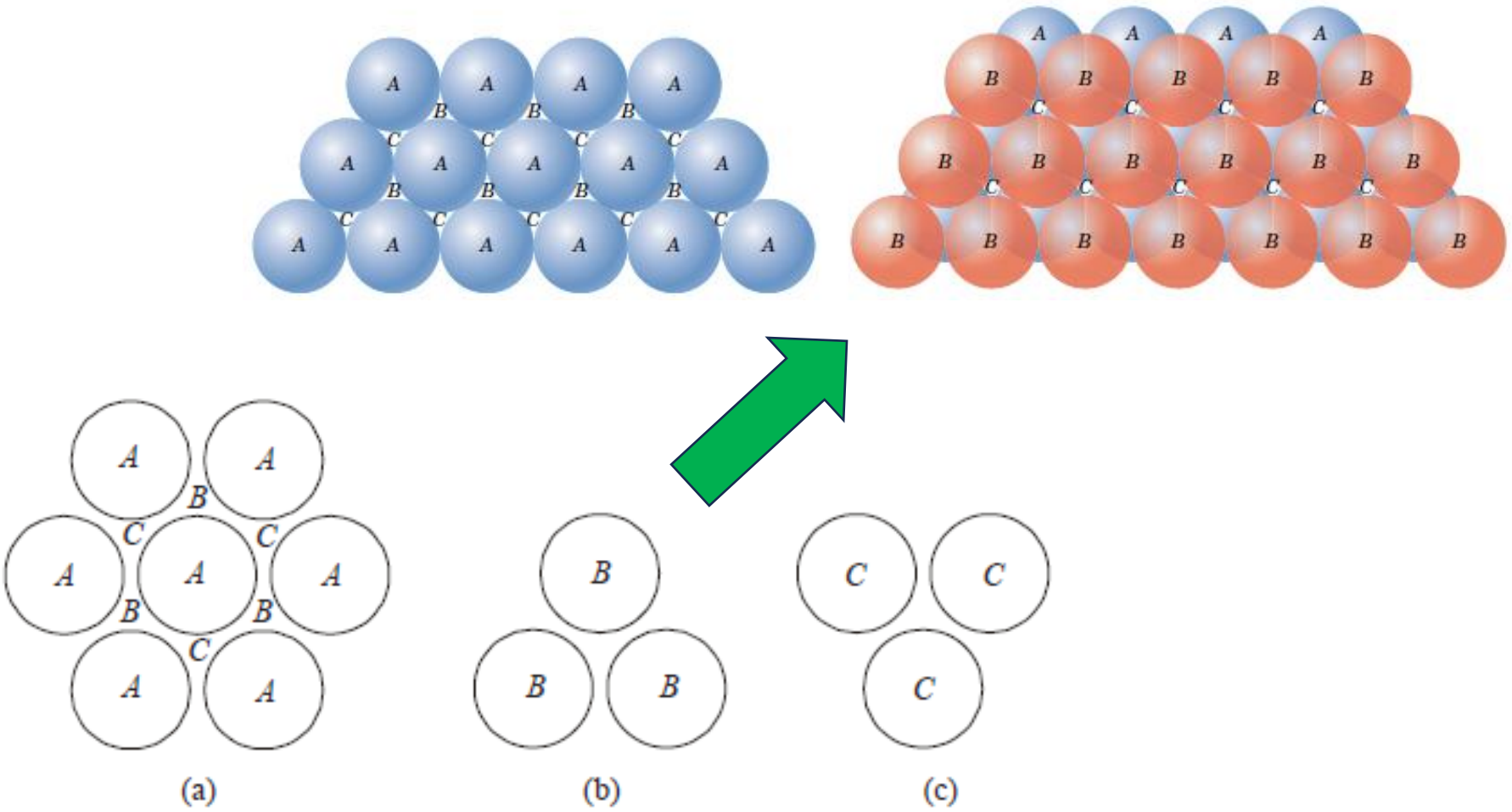
Copper

$$\rho = \frac{nA_{Cu}}{V_C N_A} = \frac{nA_{Cu}}{(16R^3\sqrt{2})N_A}$$

$$= \frac{(4 \text{ atoms/unit cell})(63.5 \text{ g/mol})}{[16\sqrt{2}(1.28 \times 10^{-8} \text{ cm})^3/\text{unit cell}](6.022 \times 10^{23} \text{ atoms/mol})}$$

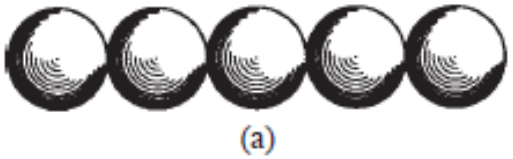
$$= 8.89 \text{ g/cm}^3$$

Close packed structures

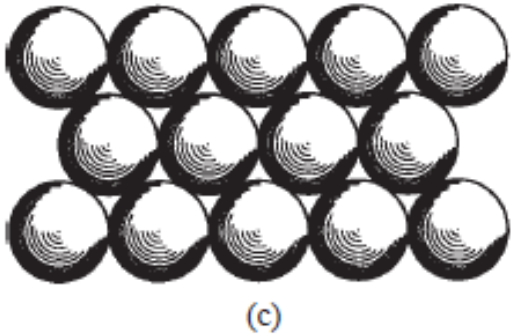


(a) A central sphere surrounded by six spheres on a close packed plane forms two sets of alternate valleys BBB and CCC; (b) three spheres to fit in the set of alternate valleys BBB; (c) three spheres to fit in the set of alternate valleys CCC

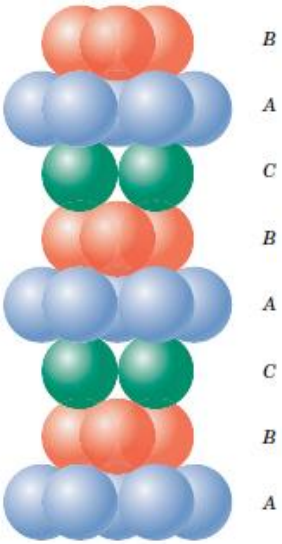
Close packed structures

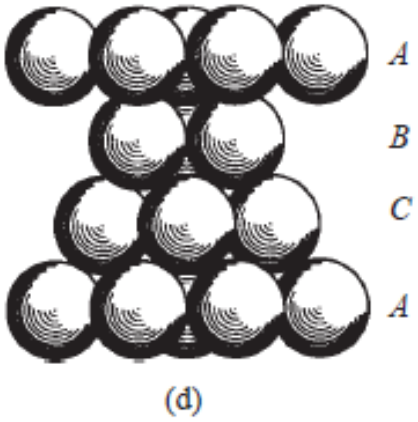


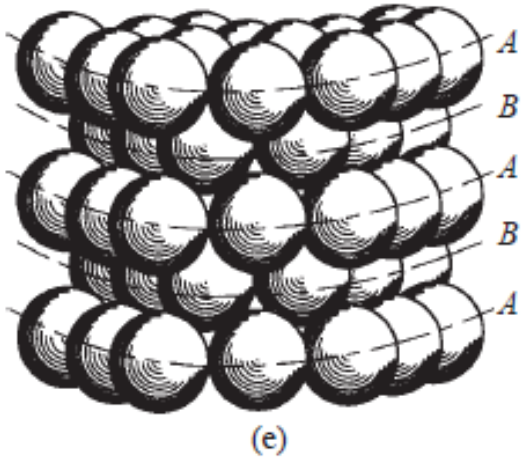




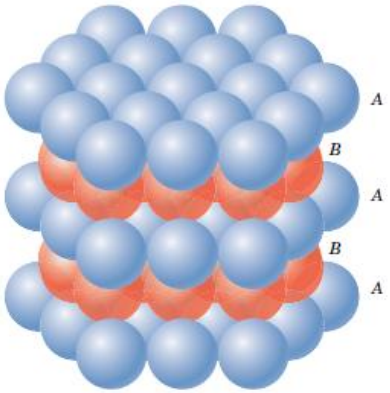
FCC





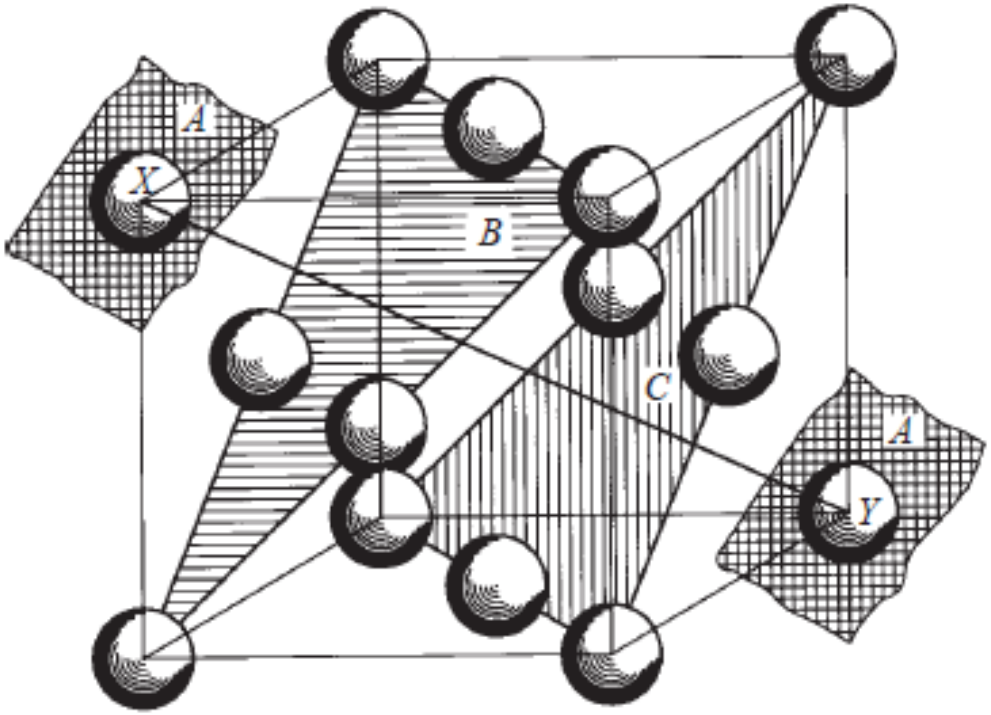


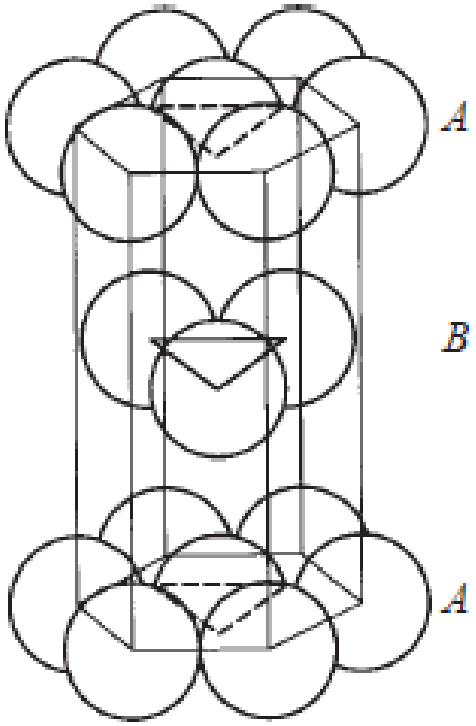
HCP



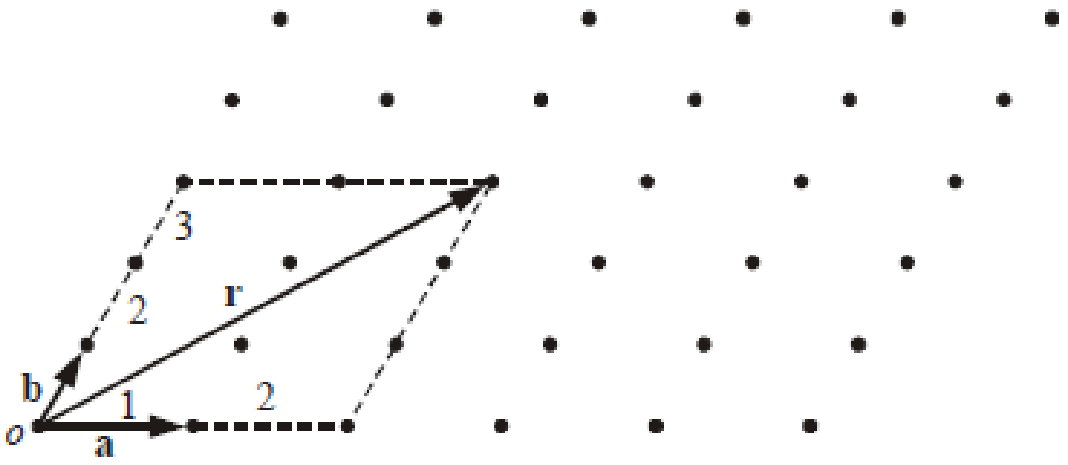
Close packing of spheres (a) along a row, (b) on two adjacent rows, (c) on a plane, (d) three-dimensional …ABCABC… stacking, and (e) …ABABA… stacking

Close packed planes in FCC and HCP unit cells

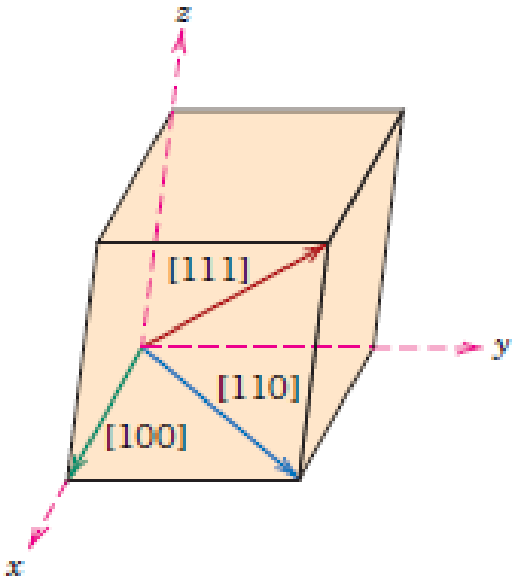




Crystal directions and planes



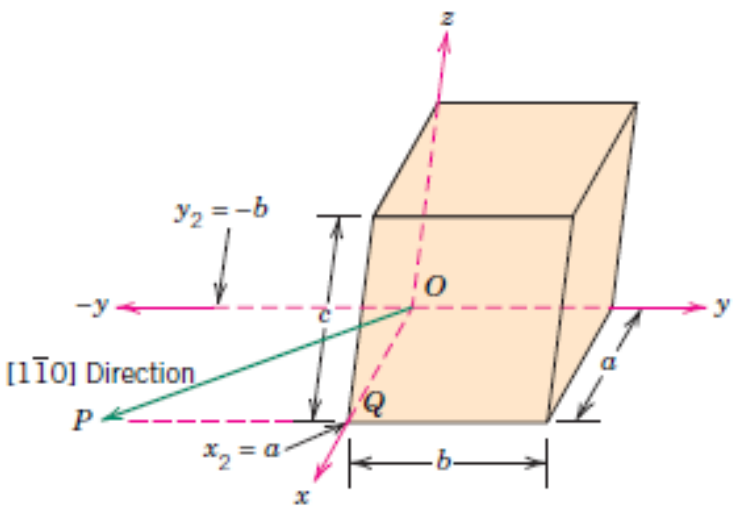
$$\mathbf{r} = r_1\mathbf{a} + r_2\mathbf{b} + r_3\mathbf{c}$$

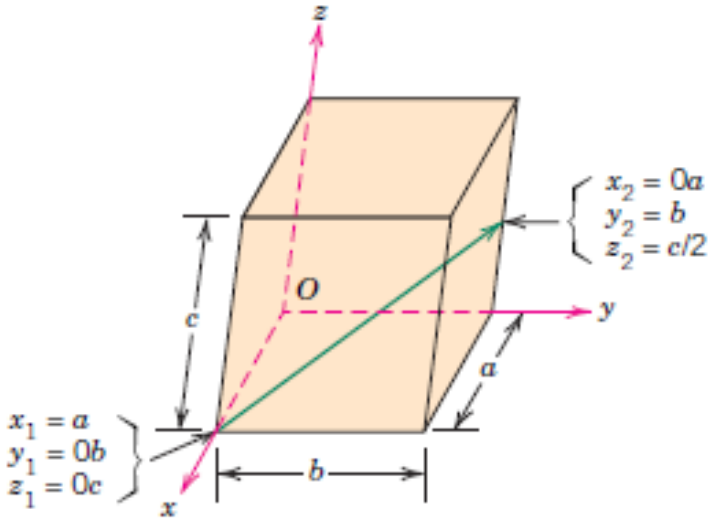


The Miller indices of the crystal direction denoted by vector r are [230].

Direction	Miller indices	Number in the family
Cube edge	⟨100⟩	6
Face diagonal	⟨110⟩	12
Body diagonal	⟨111⟩	8

Crystal directions and planes



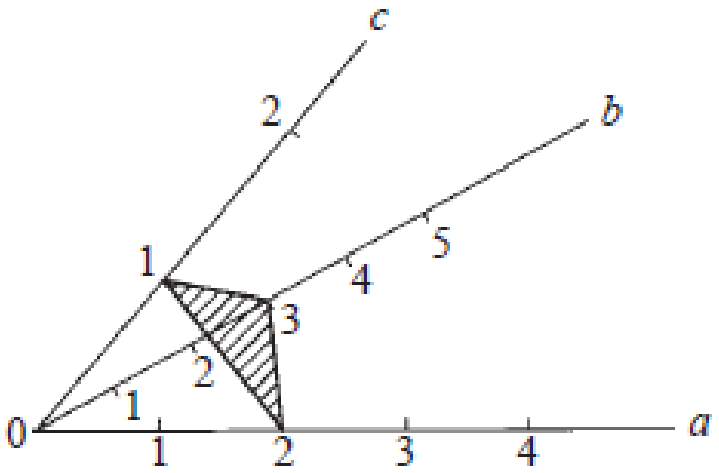


	x	y	z
Head coordinates (x_2, y_2, z_2,)	$0a$	b	$c/2$
Tail coordinates (x_1, y_1, z_1,)	a	$0b$	$0c$
Coordinate differences	$-a$	b	$c/2$
Calculated values of u, v, and w	$u = -2$	$v = 2$	$w = 1$
Enclosure		$[\bar{2}21]$	

Crystal directions and planes

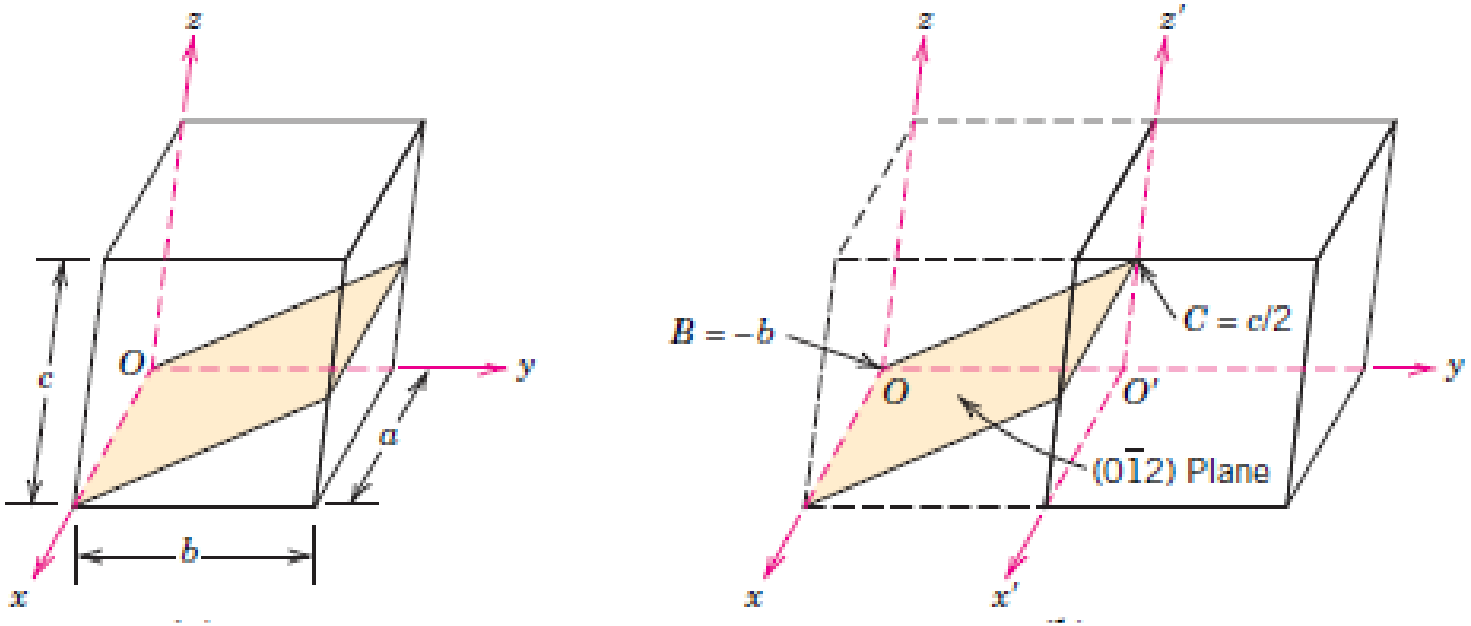
Step 1	Find the intercepts of the plane along the axes **a**, **b** and **c** (the intercepts are measured as multiples of the fundamental vectors)	2 3 1
Step 2	Take reciprocals of the intercepts[#]	$\frac{1}{2}$ $\frac{1}{3}$ 1
Step 3	Convert into smallest integers in the same ratio	3 2 6
Step 4	Enclose in *parentheses*	(326)

[#]The reciprocal procedure avoids the intercept of ∞ for a plane parallel to an axis, by making it 0.



A crystal plane making intercepts 2, 3 and 1 on the crystal axes a, b and c. Its Miller indices are (326).

Crystal directions and planes

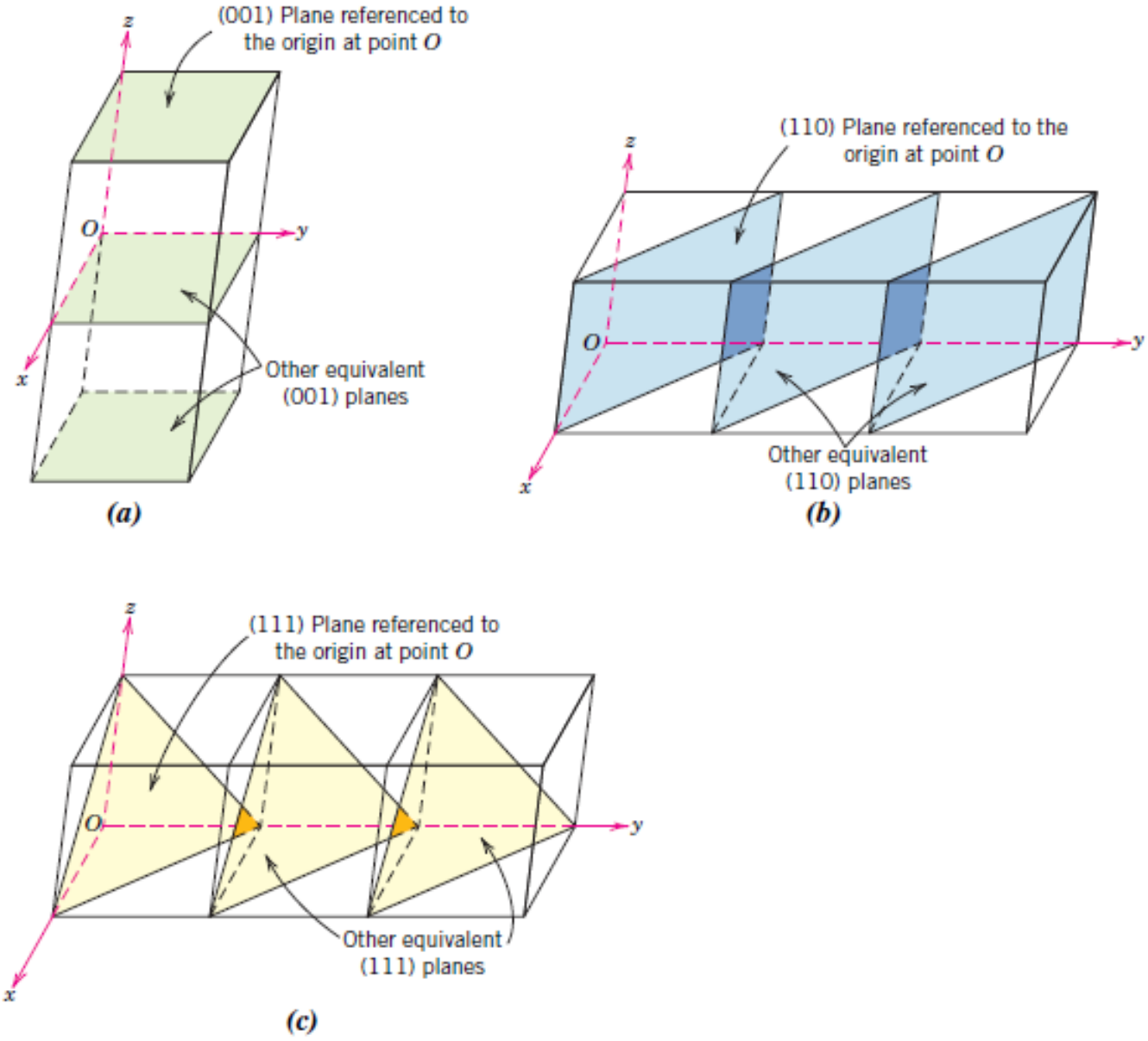


Because the plane passes through the selected origin O, a new origin must be chosen at the corner of an adjacent unit cell. In choosing this new unit cell, we move one unit-cell distance parallel to the y-axis, as shown in sketch

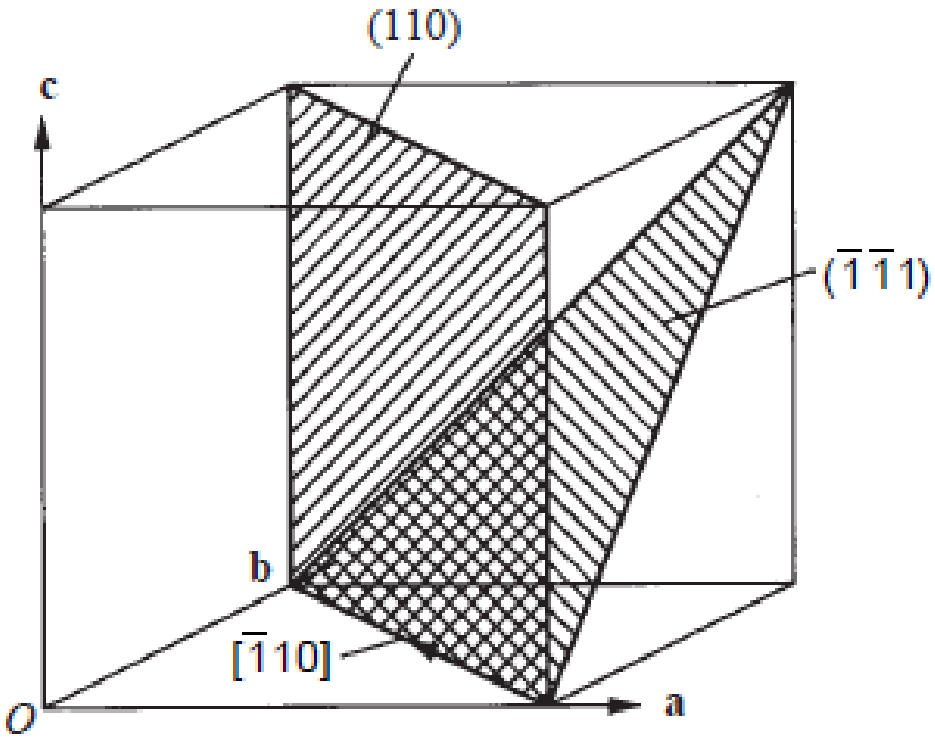
Useful conventions pertaining to Miller indices

- Unknown Miller indices are denoted by symbols h, k and l. For example, for an unknown family of directions, we write <hkl>
- When the integers used in the Miller indices contain more than one digit, the indices must be separated by commas for clarity, e.g. (3, 10, 15)
- The crystal directions of a family are not necessarily parallel to one another; similarly, not all members of a family of planes are parallel to one another
- By changing the signs of all the indices of a crystal direction, we obtain the anti-parallel or opposite direction; by changing the signs of all the indices of a plane, we obtain a plane located at the same distance on the other side of the origin
- In cubic crystals, a crystal plane and a crystal direction normal to it have the same indices, e.g., $[11\bar{1}] \perp (11\bar{1})$

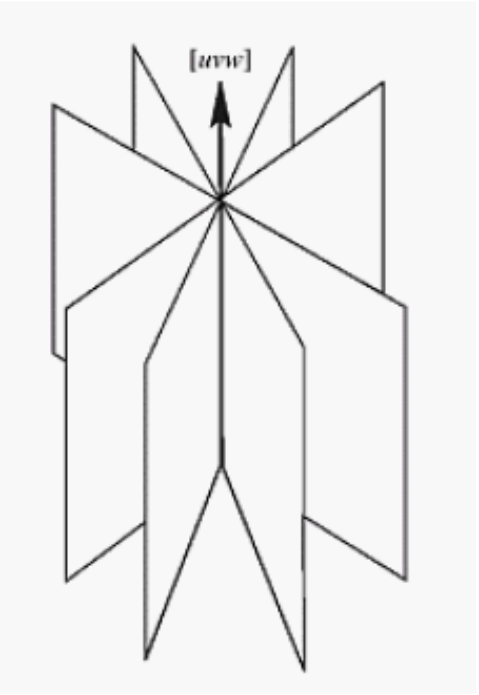
Crystal directions and planes



Direction common to two planes



Planes of a zone



$$hu + kv + lw = 0.$$

Miller-Bravais Indices

$[UVW] \rightarrow [uvtw]$

$$u = \frac{1}{3}(2U - V)$$

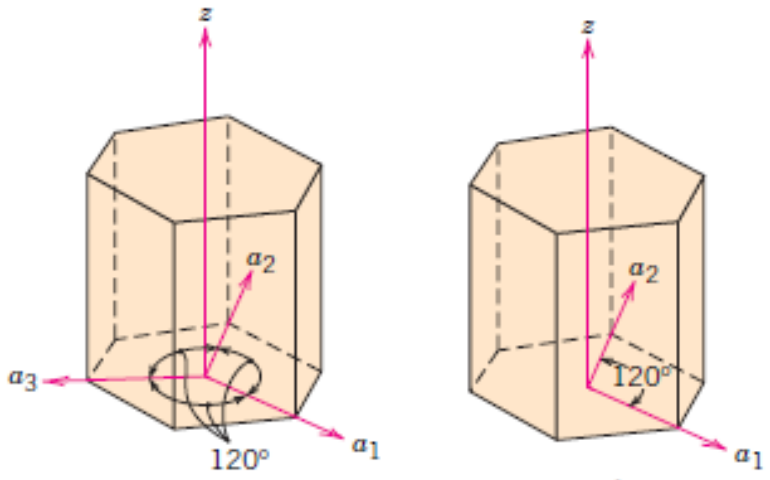
$$v = \frac{1}{3}(2V - U)$$

$$t = -(u + v)$$

$$w = W$$

Similarly for planes

$$i = -(h + k)$$



Using $a_3 = -(a_1 + a_2)$ and $t = -(u + v)$

$$u\mathbf{a}_1 + v\mathbf{a}_2 + t\mathbf{a}_3 + w\mathbf{c} = u'\mathbf{a}_1 + v'\mathbf{a}_2 + w'\mathbf{c};$$

$$u\mathbf{a}_1 + v\mathbf{a}_2 + (u+v)(\mathbf{a}_1 + \mathbf{a}_2) + w\mathbf{c} = u'\mathbf{a}_1 + v'\mathbf{a}_2 + w'\mathbf{c}.$$

from which follows:

$$u' = 2u + v;$$

$$v' = 2v + u;$$

$$w' = w.$$

The inverse relations are given by:

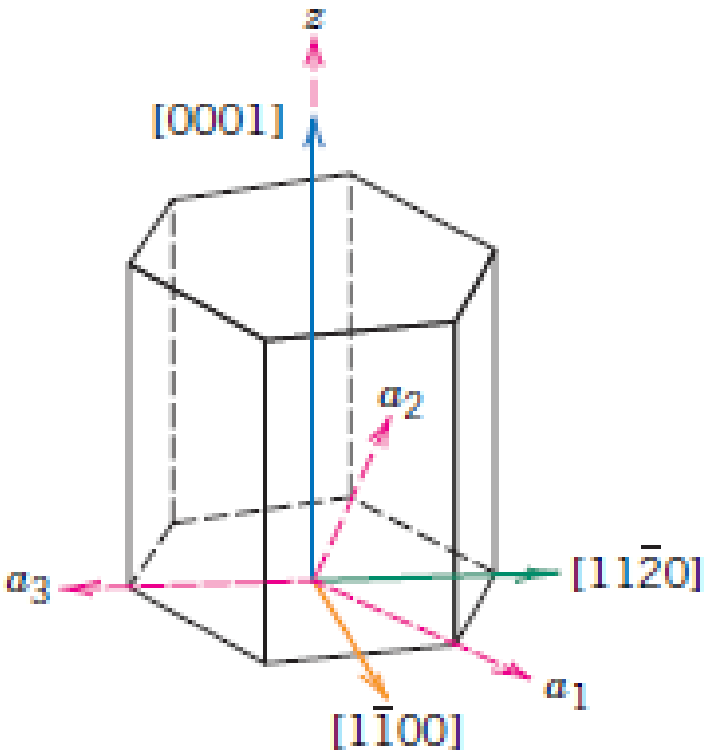
$$u = \frac{1}{3}(2u' - v');$$

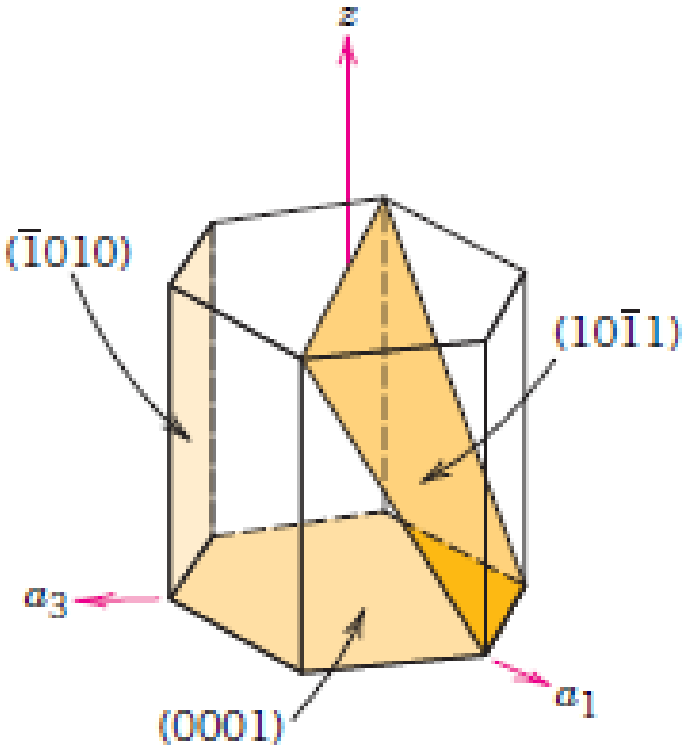
$$v = \frac{1}{3}(2v' - u');$$

$$t = -\frac{1}{3}(u' + v') = -(u + v);$$

$$w = w'.$$

Hexagonal crystal system directions and planes





References

- Materials Science and Engineering, A First Course by V. Raghavan
- Fundamentals of Materials Science and Engineering, W.D. Callister
- Structure of Materials by Marc De Graef and Michael E. McHenry